

Experiment No.	8
Experiment Title	Clustering Human Activity Recognition Data using K-Means, DBSCAN and Hierarchical Clustering ¹
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1 Aim and Objective

To implement and analyse three clustering algorithms on the Human Activity Recognition (HAR) dataset:

- **Model A:** K-Means clustering, with k selected by the Elbow method and silhouette analysis;
- **Model B:** DBSCAN (Density-Based Spatial Clustering of Applications with Noise), with ϵ and $minPts$ tuned;
- **Model C:** Hierarchical Agglomerative Clustering (HAC), with a dendrogram and a comparison of linkage criteria.

The clusters produced by each algorithm are visualised in 2-D, compared against the ground-truth activity labels (which the algorithms never see), and evaluated with both internal and external clustering metrics.

2 Dataset Description

Dataset Name	Human Activity Recognition Using Smartphones (UCI HAR)
Dataset Source	UCI Machine Learning Repository (dataset 240), retrieved via OpenML (data_id=1478) and stored as <code>har_dataset.zip</code>
Number of Samples	10,299 windows (train and test partitions pooled, since clustering is unsupervised)
Number of Features	561 engineered time- and frequency-domain features
Number of Classes	6 activities (used only for external validation)
Subjects	30 volunteers aged 19–48
Sensors	3-axis accelerometer and gyroscope at 50 Hz, windowed at 2.56s (128 readings/window) with 50% overlap
Missing Values	0 (verified programmatically)

Class balance: LAYING 1,944, STANDING 1,906, SITTING 1,777, WALKING 1,722, WALKING_UPSTAIRS 1,544, WALKING_DOWNSTAIRS 1,406 – mildly imbalanced (a 1.38:1 ratio between the largest and smallest activity), which matters for clustering because algorithms that favour equally sized clusters are at a slight disadvantage.

3 Preprocessing Steps

- **Missing value handling:** none required; the published feature matrix has no missing entries.
- **Categorical encoding:** none required for the features (all 561 are continuous). Activity names are mapped to integer indices purely so that the external metrics (ARI, NMI) can be computed.
- **Standardization:** the published features are already bounded to $[-1, 1]$, but their spreads differ, so every feature was standardized to zero mean and unit variance with `StandardScaler`. All three algorithms here are distance-based, so an unstandardized feature with a wider spread would silently dominate every distance computation.
- **Dimensionality reduction:** PCA is used for two distinct purposes – 2-D projections for the figures, and a 10-component space in which DBSCAN is tuned (Section 7 explains why DBSCAN needs this). K-Means and HAC are run on the full 561-dimensional standardized matrix.

PCA needs **104 of 561 components** to retain 95% of the variance, but the first component alone already explains **50.7%** – a strong sign that one dominant axis (body motion intensity) separates most of the data.

4 Exploratory Data Analysis

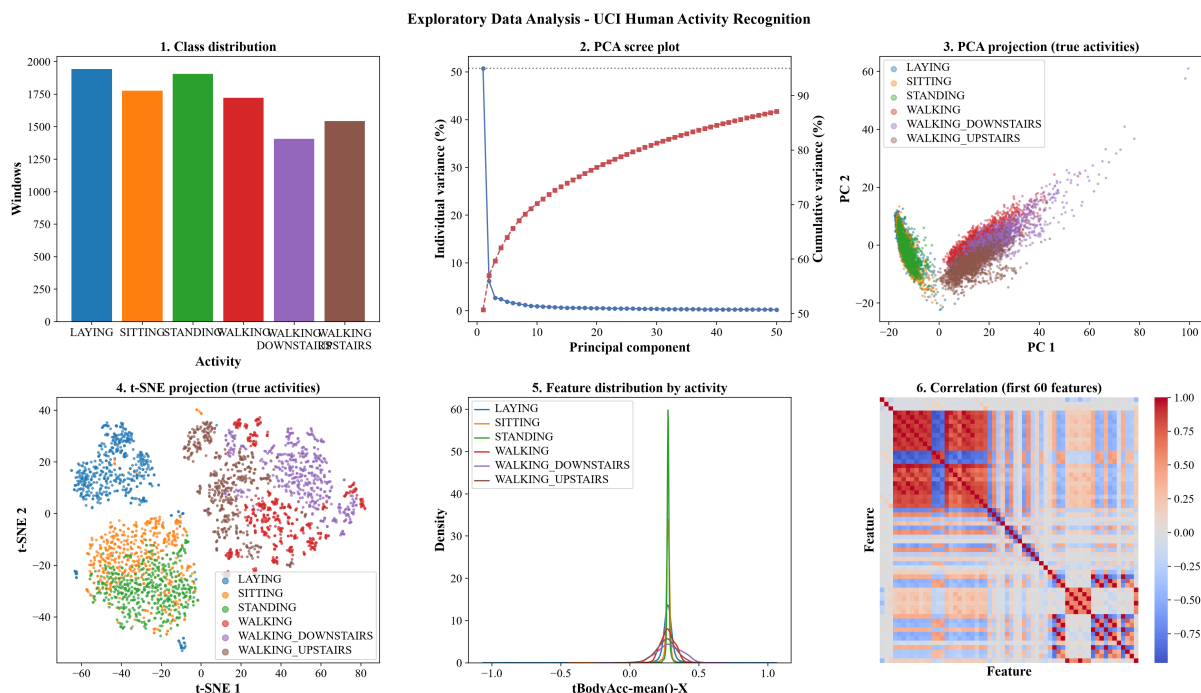


Figure 1: EDA summary: class distribution, PCA scree plot, PCA and t-SNE projections coloured by true activity, a per-activity feature distribution, and the correlation structure of the first 60 features.

Inference: The PCA projection (panel 3) shows the single most important structural fact about this dataset: the six activities form *two* widely separated super-clusters – the

three static postures (LAYING, SITTING, STANDING) and the three dynamic gaits (WALKING, WALKING_UPSTAIRS, WALKING_DOWNSTAIRS) – with a large empty gap between them. Within each super-cluster the individual activities overlap heavily: SITTING and STANDING are nearly superimposed, and the three walking variants form one elongated band. The t-SNE view (panel 4) separates LAYING cleanly and teases the walking variants apart a little, but still leaves SITTING and STANDING interleaved. This two-level structure predicts the central result of the whole experiment: unsupervised methods will find the 2-cluster split easily and the 6-activity split only partially.

5 Experimental Setup

Python Version	3.11.15
Scikit-Learn Version	1.9.0
NumPy / Pandas Version	2.4.6 / 3.0.5
SciPy Version	1.17.1
IDE	Jupyter Notebook / VS Code
Operating System	Windows 11
Hardware	Standard lab workstation (CPU only)

Environment activation command (as mandated by the lab manual):

```
1 source /opt/anaconda3/bin/activate anaconda-navigation
```

6 Model A: K-Means and the Elbow Method

Table 1: K-Means Elbow Method Results (as required by the lab manual)

k	WCSS (Inertia)	Silhouette	Davies–Bouldin	Calinski–Harabasz	ARI	NMI
2	3,272,857	0.3946	1.071	7880.8	0.3296	0.5455
3	2,921,078	0.3137	1.787	5034.5	0.3317	0.5183
4	2,781,602	0.1524	2.342	3696.3	0.2990	0.4694
5	2,654,790	0.1321	2.411	3027.3	0.2912	0.4607
6	2,577,180	0.1095	2.384	2556.5	0.4196	0.5593
7	2,519,774	0.0808	2.675	2217.9	0.4332	0.5546
8	2,465,379	0.0759	2.611	1975.2	0.4266	0.5516

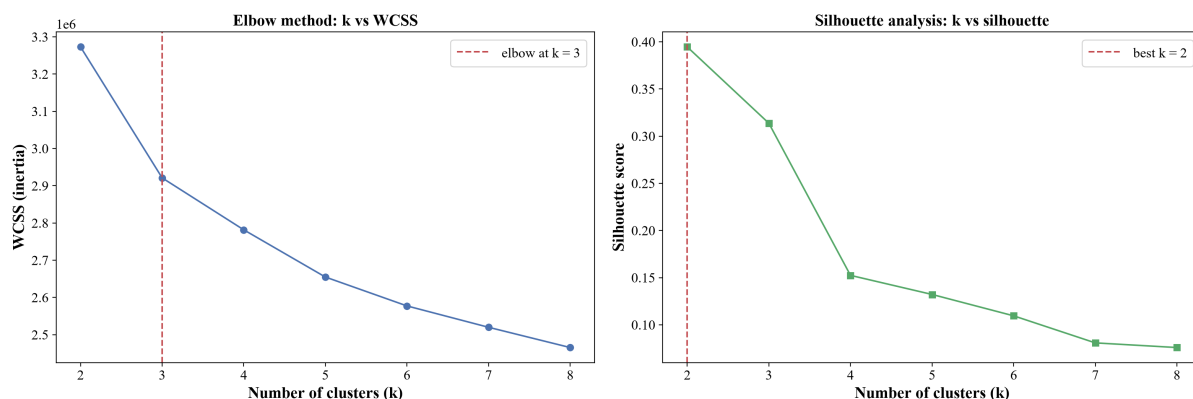


Figure 2: Left: Elbow curve (k vs WCSS). Right: silhouette score vs k .

Selecting and justifying k . The two criteria disagree, and the disagreement is itself the finding:

- **Silhouette and the elbow both point to a small k .** The silhouette score is maximised at $k = 2$ (0.3946) and falls monotonically afterwards; the sharpest bend in the WCSS curve (largest second difference) is at $k = 3$. Every internal metric agrees: Davies–Bouldin is lowest (best) and Calinski–Harabasz highest (best) at $k = 2$.
- **External agreement peaks near $k = 6$ – 7 .** ARI is maximised at $k = 7$ (0.4332) with $k = 6$ almost identical (0.4196), and NMI peaks exactly at $k = 6$ (0.5593).

$k = 2$ is the honest unsupervised answer – it is what the geometry of the data actually supports – while $k = 6$ is the answer that matches the labelling. Both solutions are therefore carried forward.

Table 2: What the $k = 2$ Solution Actually Found

True activity	Cluster 0	Cluster 1
LAYING	1932	12
SITTING	1774	3
STANDING	1906	0
WALKING	0	1722
WALKING_DOWNSTAIRS	0	1406
WALKING_UPSTAIRS	8	1536

Inference: The $k = 2$ solution is an almost perfect static-vs-dynamic separation: only 23 of 10,299 windows (0.22%) fall on the wrong side. This is why its silhouette is so much higher than any other k – it is cutting along the one genuinely wide gap in the feature space. Its ARI is only 0.3296 purely because it is being scored against a 6-class ground truth that it was never trying to reproduce.

7 Model B: DBSCAN

Why DBSCAN is run on a PCA-reduced space. DBSCAN was first run on the full 561-dimensional standardized matrix. The results demonstrate the curse of dimensionality directly: at $\epsilon = 5.0$ and $\epsilon = 7.5$ every single point is labelled noise (no point has 10 neighbours within the radius), and even at $\epsilon = 12.5$ the best result still discards 44.5% of

the data as noise with ARI 0.2207. In 561 dimensions all pairwise distances concentrate into a narrow band, so no single global radius can distinguish dense regions from sparse ones. The tuning grid was therefore repeated on the 10-component PCA projection.

Table 3: DBSCAN Tuning on the 10-D PCA Space (selected rows)

ϵ	$minPts$	Clusters	Noise Fraction	ARI	NMI
1.5	5	25	88.1%	0.0088	0.1297
2.5	5	53	52.6%	0.1857	0.3309
3.0	5	45	38.6%	0.2399	0.3880
4.0	5	30	18.1%	0.3082	0.4558
5.0	5	13	7.2%	0.3221	0.4992
6.0	5	7	3.2%	0.3274	0.5245
6.0	10	4	5.0%	0.3249	0.5150
6.0	20	3	8.1%	0.3210	0.4996
8.0	5	1	1.1%	0.0013	0.0118

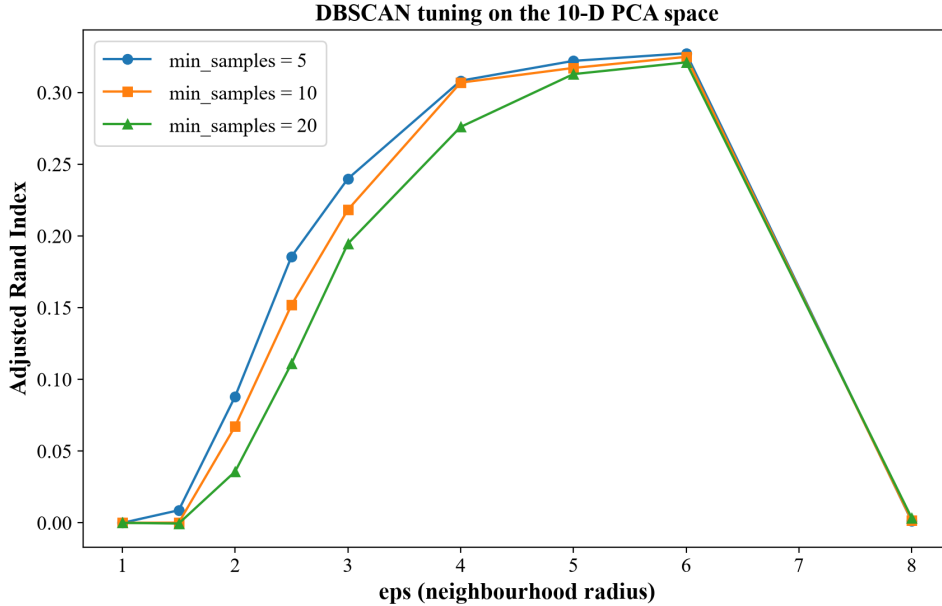


Figure 3: DBSCAN tuning: Adjusted Rand Index vs ϵ for three $minPts$ settings on the 10-D PCA space.

Inference: DBSCAN’s behaviour is dominated by ϵ , and the sweep brackets the optimum cleanly. Below $\epsilon = 2.5$ the radius is smaller than the typical nearest-neighbour distance, so 50–100% of the data is discarded as noise and the few surviving clusters are fragments. Raising ϵ steadily merges those fragments and recovers structure, peaking at $\epsilon = 6.0$, $minPts = 5$, which labels only 3.2% of windows as noise and reaches ARI 0.3274 / NMI 0.5245 with 7 clusters – close to the 6 real activities. Past that point the radius is wide enough to bridge the static/dynamic gap and the whole dataset collapses into a *single* cluster at $\epsilon = 8.0$ (ARI 0.0013). The $minPts$ parameter acts as a smoother rather than a driver: at fixed $\epsilon = 6.0$, raising it from 5 to 20 merely merges the smaller clusters ($7 \rightarrow 3$) and raises the noise fraction from 3.2% to 8.1%, with almost no change in ARI.

8 Model C: Hierarchical Agglomerative Clustering

Ward linkage requires the full pairwise distance structure, so the linkage comparison and the dendrogram are computed on a stratified subsample of 3,000 windows (500 per activity). The same subsample is used for every linkage, so the rows of Table 4 are directly comparable.

Table 4: Linkage Criterion Comparison (HAC, $k = 6$, 3,000-window subsample)

Linkage	Silhouette	Davies–Bouldin	Calinski–Harabasz	ARI	NMI
Ward	0.0809	2.726	697.6	0.3434	0.4946
Complete	0.3851	1.069	63.1	0.0009	0.0290
Average	0.4193	1.155	63.6	0.0009	0.0294
Single	0.5542	0.267	17.4	0.0000	0.0033

Inference: This table is the clearest cautionary tale in the experiment: **the linkage with the best internal scores is the worst clustering**. Single linkage achieves a silhouette of 0.5542 and a Davies–Bouldin of 0.267 – numbers that look excellent in isolation – while scoring an ARI of essentially zero. It does this by the classic single-linkage failure mode: chaining almost every point into one giant cluster and splitting off a handful of outliers as the remaining five. Complete and average linkage fail the same way, less extremely. Only Ward linkage, which explicitly minimises within-cluster variance and therefore resists chaining, produces a clustering that corresponds to the real activities (ARI 0.3434), and it does so while posting the *lowest* silhouette of the four.

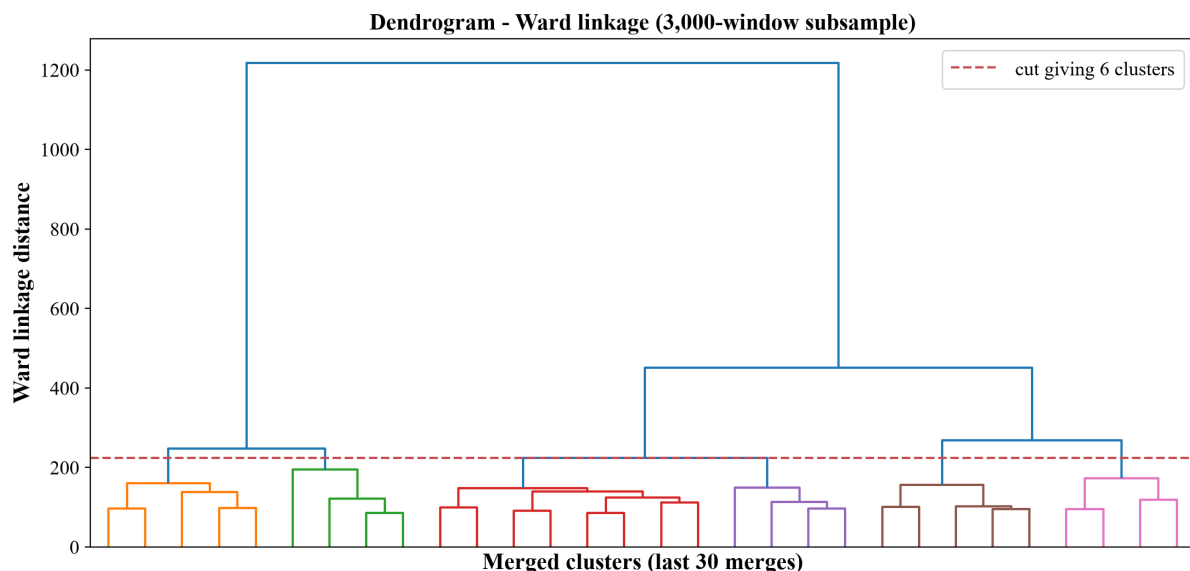


Figure 4: Dendrogram (Ward linkage, last 30 merges, 3,000-window subsample).

Inference: The dendrogram’s two highest merges are separated from everything below them by a large vertical gap – the same static/dynamic split K-Means found at $k = 2$. Cutting below that gap yields the 6-cluster solution used in Table 5.

9 Comparison of All Three Algorithms

Table 5: Clustering Algorithm Comparison

Algorithm	Clusters	Noise	Silhouette	Davies–Bouldin	Calinski–Harabasz	ARI	NMI
K-Means ($k = 2$)	2	0%	0.3946	1.071	7880.8	0.3296	0.5455
K-Means ($k = 6$)	6	0%	0.1095	2.384	2556.5	0.4196	0.5593
DBSCAN ($\epsilon = 6.0$, $minPts = 5$)	7	3.2%	0.4151	0.770	3449.5	0.3274	0.5245
Hierarchical (Ward, $k = 6$)	6	0%	0.0809	2.726	697.6	0.3434	0.4946

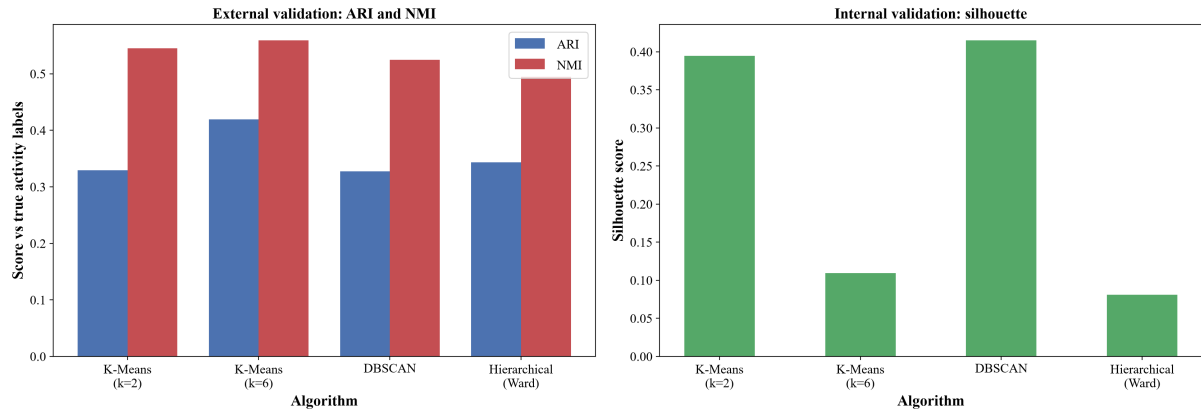


Figure 5: External validation (ARI, NMI) and internal validation (silhouette) for all four clustering solutions.

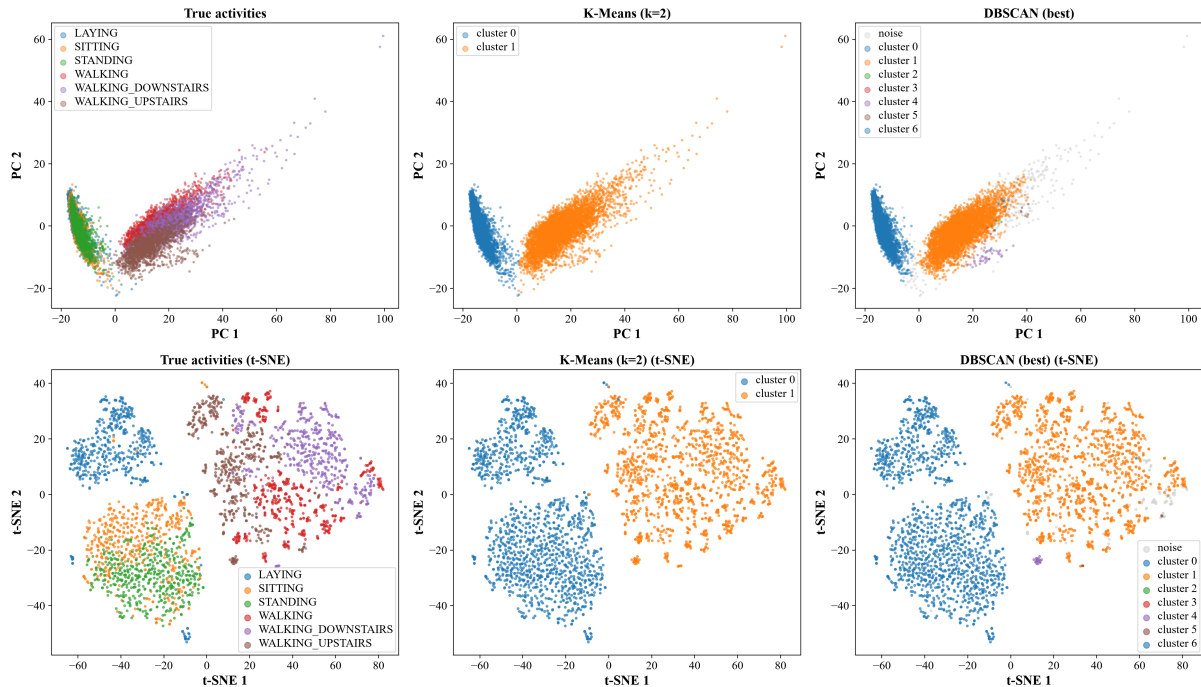


Figure 6: Cluster assignments in 2-D. Top row: PCA projection. Bottom row: t-SNE projection. Left to right: true activities, K-Means, DBSCAN.

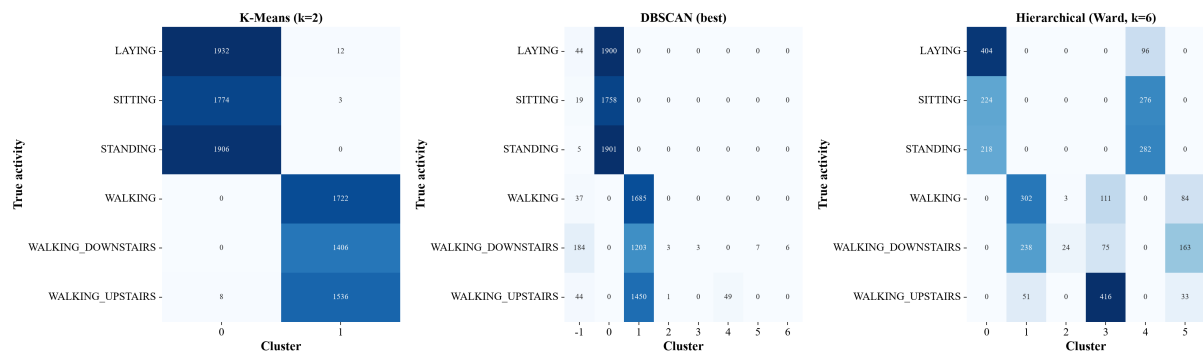


Figure 7: Cluster-to-activity contingency tables for the three algorithms.

10 Observation Questions

- Which algorithm produced the most meaningful clusters? Why?** It depends on what "meaningful" is measured against, and the two answers are genuinely different. **Against the activity labels, K-Means with $k = 6$ wins** (ARI 0.4196, NMI 0.5593) – no other configuration matches the labelling as closely. **Against the geometry of the data, the 2-cluster solutions win:** the $k = 2$ K-Means split separates static from dynamic activities with 99.8% purity and posts the best Calinski–Harabasz score by a factor of two. DBSCAN at $\epsilon = 6.0$ is arguably the best all-round compromise: it finds 7 clusters without being told how many, keeps 96.8% of the data, and achieves both the best silhouette (0.4151) and the best Davies–Bouldin (0.770) of any solution while staying competitive on NMI (0.5245). The honest summary is that all three algorithms find the same dominant structure – motion versus stillness – and all three struggle with the same boundary, SITTING versus STANDING, which is nearly invisible to accelerometer features because both are motionless postures differing only in orientation.
- How sensitive was K-Means to the choice of k ?** Very sensitive, and in opposite directions depending on the metric. Internal quality collapses as k grows: the silhouette falls from 0.3946 at $k = 2$ to 0.0759 at $k = 8$, a 5-fold decrease, because each additional split has to cut through a genuinely dense region. External agreement does the opposite, and non-monotonically: ARI dips from 0.3296 ($k = 2$) to 0.2912 ($k = 5$), then jumps sharply to 0.4196 at $k = 6$. That jump is the signature of $k = 6$ finally having enough clusters to place one in each activity region. So the choice of k does not merely change cluster granularity, it changes which structure is recovered – and the elbow and silhouette heuristics alone would have led to $k = 2$ or $k = 3$, neither of which recovers the activities.
- Did DBSCAN detect noise or small clusters effectively?** Yes, once given a space it could work in. The noise fraction is a direct, tunable dial: 88.1% at $\epsilon = 1.5$ falling to 3.2% at $\epsilon = 6.0$. At the chosen setting DBSCAN isolates 333 windows (3.2%) as noise while keeping 7 clusters, and those clusters earn the highest silhouette of any method here – evidence that excluding the sparse boundary points genuinely sharpens the remaining structure, which is exactly what DBSCAN is for and what neither K-Means nor HAC can do (both are forced to assign every point). It also found small clusters that the partitional methods cannot express: the $\epsilon = 5.0$

solution splits into 13 clusters, several of them small sub-groups within the dynamic activities. The important qualification is that **DBSCAN completely failed on the raw 561-dimensional data** – 100% noise at $\epsilon \leq 7.5$, and 44.5% noise at best – so its success is conditional on the PCA reduction, not intrinsic.

- **How does linkage choice (single/complete/ward) affect hierarchical clustering?** Dramatically – it is the difference between a working clustering and a useless one. Ward linkage achieves ARI 0.3434, while complete (0.0009), average (0.0009) and single (0.0000) linkage all produce essentially random labellings. Single, complete and average linkage merge on extremal point-to-point distances, so on a dataset with a dense core and a sparse fringe they chain nearly everything into one cluster and split off tiny outlier groups. Ward instead merges whichever pair minimises the increase in total within-cluster variance, a criterion that resists chaining and prefers balanced, compact clusters – which matches how these activities actually sit in feature space.
- **Which internal metric best matched your visual intuition of cluster quality?** None of them matched it reliably, and that is the most useful lesson in this experiment. **Silhouette was actively misleading in the linkage comparison:** single linkage scored 0.5542, the best of all four linkages, for a clustering that is meaningless (ARI 0.0000), while Ward scored the worst (0.0809) for the only usable one. Davies–Bouldin agreed with silhouette and was equally wrong (0.267 for single linkage). The one internal metric that tracked visual intuition most closely was **Calinski–Harabasz**, which ranked Ward (697.6) an order of magnitude above single linkage (17.4) and correctly preferred the visually obvious 2-cluster split (7880.8). The general pattern is that silhouette and Davies–Bouldin reward compact, well-separated clusters *whatever those clusters contain* – so one giant cluster plus a few tight outlier groups can score beautifully. Internal metrics are best read alongside a projection plot and, where labels exist, alongside ARI/NMI.

11 Conclusion

Three clustering algorithms were applied to 10,299 windows of smartphone sensor data described by 561 engineered features, with the six activity labels held back and used only for external validation.

All three recover the same dominant structure, and it is not the six-activity structure the labels describe: the data’s strongest signal is a two-way split between static postures (LAYING, SITTING, STANDING) and dynamic gaits (WALKING, WALKING_UPSTAIRS, WALKING_DOWNSTAIRS). K-Means finds that split at $k = 2$ with 99.8% purity, the Ward dendrogram shows it as the highest and widest merge, and DBSCAN’s collapse into one cluster at $\epsilon = 8.0$ marks the exact radius at which the gap is bridged. Measured against the labels, K-Means at $k = 6$ is the best performer (ARI 0.4196, NMI 0.5593); measured on internal quality, DBSCAN at $\epsilon = 6.0$ is (silhouette 0.4151, Davies–Bouldin 0.770, with only 3.2% of points discarded as noise).

The residual errors are consistent across every method and have a physical explanation: the $k = 6$ contingency table shows SITTING and STANDING merged into one cluster (1,238 and 1,346 windows in cluster 0), because both are motionless and

accelerometer-derived features cannot separate two stationary postures that differ only in trunk orientation. The walking variants, by contrast, are separated reasonably well.

Two methodological findings are worth carrying forward. First, DBSCAN is unusable on the raw 561-dimensional space – every point is labelled noise at small radii because distances concentrate in high dimensions – and only becomes competitive after PCA reduction to 10 components. Second, internal validation metrics can be actively misleading: single linkage posted the best silhouette and Davies–Bouldin scores in the entire experiment while producing a clustering with an ARI of zero. Cluster quality should be judged from projections and, when labels exist, external metrics – not from a single internal score.

12 References

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